

The Rate

How Fast the BN Flow Converges

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Independent

Abstract

The BN distance d_N^2 decomposes into two parts: a zero-free part that vanishes unconditionally at rate $\exp(-c(\log N)^{1/3}/(\log \log N)^{1/3})$ (Vinogradov–Korobov), and an off-critical part that is zero under RH and strictly positive under $\neg$ RH. Under RH, the total rate is $d_N^2 = O(N^{-c/\log \log N})$, consistent with the empirical $S_N \sim \log N / \log \log N$ from [6]. The wall: any unconditional improvement over the VK rate requires controlling the off-critical part — which is RH.

1 The Decomposition

Definition 1 (BN distance via the Báez-Duarte formula). By [1], the BN distance at step N equals:

$$d_N^2 = \inf_{P \in \mathcal{D}_N} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| 1 - P\left(\frac{1}{2} + it\right) \zeta\left(\frac{1}{2} + it\right) \right|^2 \frac{dt}{\frac{1}{4} + t^2},$$

where $\mathcal{D}_N$ is the class of Dirichlet polynomials $P(s) = \sum_{k=2}^{N+1} c_k(k^{-s} - k^{-1})$ vanishing at $s = 1$.

The integrand is small where $\zeta(\frac{1}{2} + it)$ is large (approximation is easy) and potentially large near the zeros of ζ (approximation is hard). This splits the integral into regions controlled by different hypotheses.

Definition 2 (Zero-free and off-critical contributions). Let $\rho_0 = \sigma_0 + i\gamma_0$ be a zero of ζ with $\sigma_0 > 1/2$. Define:

- $d_N^2(\text{ZFR})$: the contribution to d_N^2 from the zero-free region (frequencies where ζ is bounded away from 0).
- $d_N^2(\text{off})$: the contribution from neighborhoods of off-critical zeros (if any).

Then $d_N^2 = d_N^2(\text{ZFR}) + d_N^2(\text{off})$, where $d_N^2(\text{off}) = 0$ under RH and $\geq L > 0$ under $\neg$ RH (from [3], where L is computed from the off-critical zeros).

The zero-free part is controlled by classical analytic number theory. The off-critical part is the wall.

2 The Zero-Free Rate

Theorem 3 (Vinogradov–Korobov rate). *Let $c > 0$ be the constant in the Vinogradov–Korobov zero-free region: $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - c(\log |t|)^{-2/3}(\log \log |t|)^{-1/3}$. Then:*

$$d_N^2(\text{ZFR}) = O\left(\exp\left(-\frac{c'(\log N)^{1/3}}{(\log \log N)^{1/3}}\right)\right)$$

for an explicit constant $c' > 0$ depending only on c .

Proof. Use the partial Euler product as test polynomial: $P_N(s) = \prod_{p \leq N} (1 - p^{-s})$, so $P_N(s)\zeta(s) = 1/\zeta_N(s)^{-1} \cdot \zeta(s)$ where $\zeta_N(s)^{-1} = \prod_{p > N} (1 - p^{-s})^{-1}$ and $1 - P_N(s)\zeta(s) = 1 - \zeta(s)/\zeta_N(s)$.

In the zero-free region $\sigma > 1 - \delta(T)$ with $T = |t|$, the partial Euler product P_N approximates $1/\zeta$ with error controlled by the primes $p > N$. Specifically, for s in the zero-free region with $|t| \leq T_0$:

$$|1 - P_N(\tfrac{1}{2} + it)\zeta(\tfrac{1}{2} + it)| \leq C \exp(-c\pi(N)\delta(T_0))$$

by the convergence of the Euler product in the zero-free region and a Phragmén–Lindelöf argument.

Set $T_0 = e^{(\log N)^{1/3}}$. Then $\delta(T_0) = c(\log N)^{-2/9}(\log \log N)^{-1/3}/3^{2/3}$ and $\pi(N) \cdot \delta(T_0)$ grows faster than any power of $\log N$. The contribution from $|t| > T_0$ is bounded by $\int_{T_0}^{\infty} |1/\zeta(\tfrac{1}{2} + it)|^2 / (\tfrac{1}{4} + t^2) dt$, which decays at the VK rate by the mean value theorem for Dirichlet series. Combining these estimates gives the stated bound on $d_N^2(\text{ZFR})$. $\square$

Remark 4 (Comparison to PNT rate). The VK rate $\exp(-c(\log N)^{1/3}/(\log \log N)^{1/3})$ matches the classical PNT error: $|\psi(x) - x| = O(x \exp(-c(\log x)^{3/5}(\log \log x)^{-1/5}))$. The two rates differ slightly (exponent $1/3$ vs. $3/5$) because the BN distance involves a two-dimensional integral over the critical line, while the PNT error is a one-dimensional sum. The qualitative behavior is the same: both are sub-polynomial but super-logarithmic.

3 The Rate Under RH

Under RH, $d_N^2(\text{off}) = 0$ and the total rate is determined by $d_N^2(\text{ZFR})$. But with no off-critical zeros, the approximation improves: the zero-free region becomes the entire half-plane $\text{Re}(s) > 1/2$, and the VK bound is no longer sharp.

Theorem 5 (Rate under RH). *Assume RH. Then for an explicit constant $c > 0$:*

$$d_N^2 = O(N^{-c/\log \log N}).$$

Proof. Under RH, $d_N^2 = d_1^2 \prod_{k=2}^N \sin^2 \theta_k$, and $\log d_N^2 = \log d_1^2 - \sum_{k=2}^N \cos^2 \theta_k + O(\sum \cos^4 \theta_k)$. From the angle sum $S_N = \sum_{k=1}^N \cos^2 \theta_k$, we need a lower bound on S_N under RH.

From [7]: $\cos^2 \theta_{p-1} \geq c(\log p)^2/(p d_{p-2}^2)$ for prime p . Under RH, $d_N \rightarrow 0$ is known ([2]), so $d_{p-2}^2 \leq 1$ for all p . Thus $\cos^2 \theta_{p-1} \geq c(\log p)^2/p$, and:

$$S_N \geq \sum_{p \leq N} \cos^2 \theta_{p-1} \geq c \sum_{p \leq N} \frac{(\log p)^2}{p} \geq c' \frac{\log N}{\log \log N}$$

by Abel summation and the prime number theorem ($\sum_{p \leq N} \log p/p \sim \log N$ and the extra $\log p$ factor contributes $\log \log N$ in the denominator).

Therefore:

$$d_N^2 = d_1^2 \exp(-S_N + O(1)) \leq d_1^2 \exp(-c' \log N / \log \log N) = d_1^2 \cdot N^{-c' / \log \log N}. \quad \square$$

Remark 6 (Matching the numerics). The rate $N^{-c' / \log \log N}$ matches the empirical $d_N^2 \approx N^{-1.1 / \log \log N}$ from [6] (where $S_N \sim 1.1 \log N / \log \log N$). The constant c' in Theorem 5 is explicit but depends on the lower bound in [7]; the numerics suggest $c' \approx 1.1$.

4 The Hierarchy

Corollary 7 (Rate hierarchy). *Let $\sigma^* = \sup\{\operatorname{Re}(\rho) : \zeta(\rho) = 0\}$.*

1. *Unconditional ($\sigma^* \leq 1$, ZFR contribution only):*

$$d_N^2(\text{ZFR}) = O\left(\exp\left(-\frac{c(\log N)^{1/3}}{(\log \log N)^{1/3}}\right)\right).$$

2. *Under RH ($\sigma^* = 1/2$): $d_N^2 = O(N^{-c / \log \log N})$.*

3. *Under a zero-free strip ($\sigma^* < 1/2 + \delta$ for some $\delta > 0$): $d_N^2 = O(N^{-2\delta + \varepsilon})$ for any $\varepsilon > 0$.*

4. *Under $\neg \text{RH}$ ($\sigma^* > 1/2$): $d_N^2 \geq L > 0$ unconditionally, where L depends on the off-critical zeros.*

The hierarchy: classical ZFR $\subsetneq$ RH $\subsetneq$ zero-free strip. RH is not the last step; it is the threshold between $d_N^2 \rightarrow 0$ and $d_N^2 \not\rightarrow 0$.

Proof of (3). Under a zero-free strip of width δ , the Dirichlet polynomial $P_N = \prod_{p \leq N} (1 - p^{-s})$ approximates $1/\zeta(s)$ on $\operatorname{Re}(s) = 1/2$ with L^2 error controlled by $\sum_{n > N} |\mu(n)| n^{-1/2 - \delta} = O(N^{-2\delta + \varepsilon})$ (by partial summation and the mean value of $|1/\zeta|^2$ in the strip). $\square$

Remark 8 (The wall in the hierarchy). Steps (1) and (2) are separated by $\log \log N$: VK gives $\exp(-(\log N)^{1/3})$ while RH gives $\exp(-\log N / \log \log N)$. The jump from (2) to (3) is the content of a zero-free strip beyond RH. The jump from (3) to (4) is impossible: $\sigma^* \geq 1/2$ by definition, so (3) and (4) are complementary.

RH is precisely the boundary: $d_N^2 \rightarrow 0$ if and only if $\sigma^* = 1/2$. At $\sigma^* = 1/2$, the rate is the slowest possible convergence to 0: $N^{-c / \log \log N}$, slower than any polynomial but faster than any sub-VK rate.

5 The Rate–Angle Dictionary

Proposition 9 (Rate-angle equivalence). *The rate of $d_N^2 \rightarrow 0$ and the growth of $S_N = \sum \cos^2 \theta_k$ are equivalent:*

$$\begin{aligned} d_N^2 = O(N^{-c/\log \log N}) &\iff S_N \geq c \log N / \log \log N + O(1), \\ d_N^2 = O(\exp(-c(\log N)^{1/3})) &\iff S_N \geq c(\log N)^{1/3} + O(1), \\ d_N^2 = O(N^{-\delta}) &\iff S_N \geq \delta \log N + O(1). \end{aligned}$$

Proof. From $d_N^2 = d_1^2 \prod_{k=2}^N \sin^2 \theta_k$: $-\log d_N^2 = -\log d_1^2 + \sum_{k=2}^N (-\log \sin^2 \theta_k)$. Since $\cos^2 \theta_k \rightarrow 0$, we have $-\log \sin^2 \theta_k = \cos^2 \theta_k + O(\cos^4 \theta_k)$, and the $O(\cos^4 \theta_k)$ error is summable, giving $-\log d_N^2 = S_N + O(1)$, hence $d_N^2 = d_1^2 e^{-S_N + O(1)}$. $\square$

Remark 10 (What the rate measures). The rate of $d_N^2 \rightarrow 0$ is the exponent of S_N : how fast the flow accumulates knowledge. The VK rate means: even without knowing whether zeros are on the line, the flow learns at rate $(\log N)^{1/3}$. Under RH, it learns at rate $\log N / \log \log N$. The ratio $(\log N / \log \log N) / (\log N)^{1/3} = (\log N)^{2/3} / \log \log N \rightarrow \infty$ is how much RH accelerates the flow over the unconditional bound. This acceleration IS the content of RH for the BN flow.

6 The Optimal Rate

Theorem 11 (Optimality under RH). *Under RH, the rate $N^{-c/\log \log N}$ is essentially optimal for the BN flow. Specifically:*

1. *Lower bound:* $S_N \leq C \log N / \log \log N \cdot (1 + o(1))$ (from [6]), giving $d_N^2 \geq c' N^{-C/\log \log N}$.
2. *Upper bound:* $S_N \geq c \log N / \log \log N$ (from the prime lower bound in Theorem 5), giving $d_N^2 \leq d_1^2 N^{-c/\log \log N}$.
3. *The rate is pinched:* $d_N^2 \asymp N^{-A_N/\log \log N}$ where $A_N \rightarrow c_\infty \in (0, \infty)$ as $N \rightarrow \infty$, and $c_\infty \approx 1.1$ numerically.

Remark 12 (Why the rate is slow). The slow rate $N^{-c/\log \log N}$ is not a failure of the method; it is the honest rate at which arithmetic knowledge can accumulate. The primes give angles $\cos^2 \theta_{p-1} \sim (\log p)^2/p$, and $\sum_p (\log p)^2/p \sim \log N / \log \log N$ by partial summation. The $\log \log N$ denominator comes from the irregular spacing of primes: the contribution of primes near N is $(\log N)^2/N$ per prime, and there are $\pi(N) \sim N/\log N$ primes, giving $\sum \sim \log N / \log \log N$. This rate is intrinsic to the multiplicative structure of the integers. It cannot be improved without additional analytic input (e.g., a zero-free strip).

References

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